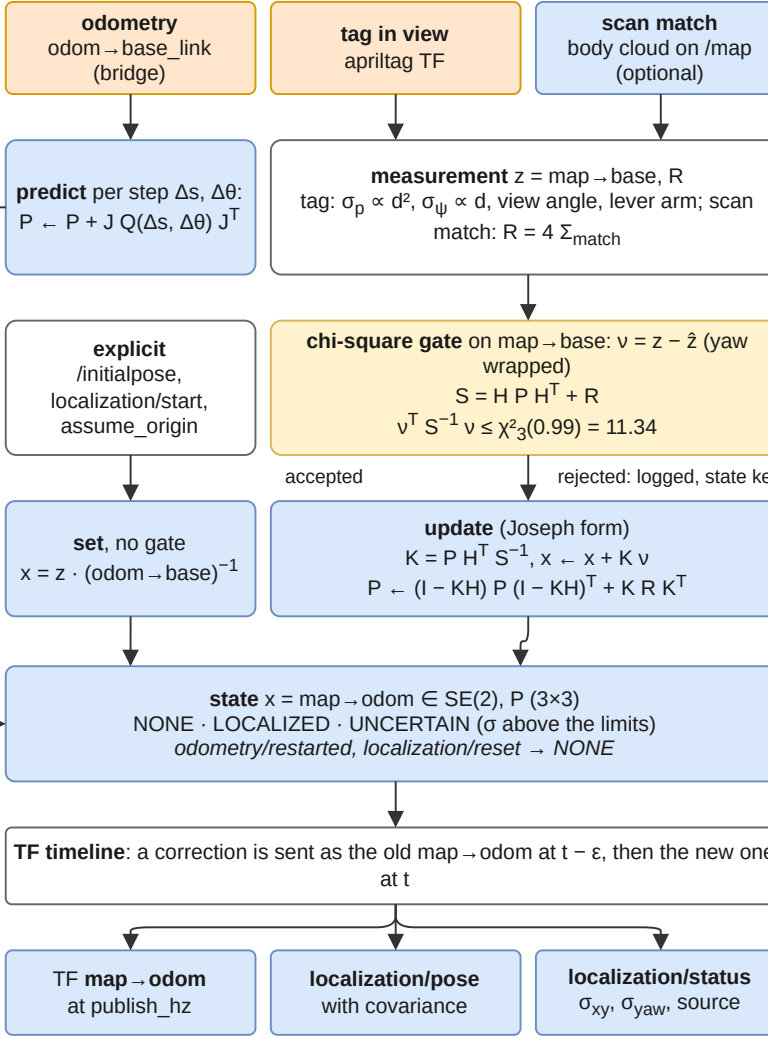
odometry
odom → base_link
(bridge)
tag in view
apriltag TF
scan match
body cloud on /map
(optional)
predict per step Δs, Δθ:
P ← P + J Q(Δs, Δθ) J^T
measurement z = map → base, R
tag: σ_p ∝ d², σ_ψ ∝ d, view angle, lever arm; scan
match: R = 4 Σ_match
explicit
/initialpose,
localization/start,
assume_origin
chi-square gate on map → base: ν = z − ẑ (yaw
wrapped)
S = H P H^T + R
ν^T S^−1 ν ≤ χ²_3(0.99) = 11.34
accepted
rejected: logged, state kept
set, no gate
x = z · (odom → base)^−1
update (Joseph form)
K = P H^T S^−1, x ← x + K ν
P ← (I − KH) P (I − KH)^T + K R K^T
state x = map → odom ∈ SE(2), P (3×3)
NONE · LOCALIZED · UNCERTAIN (σ above the limits)
odometry/restarted, localization/reset → NONE
TF timeline: a correction is sent as the old map → odom at t − ε, then the new one
at t
TF **map → odom**
at publish_hz
localization/pose
with covariance
localization/status
σ_xy, σ_yaw, source